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Magnetic Activities of GJ 1151: Flares in TESS Data and Radio Observation in uGMRT

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Abstract

GJ 1151 is a nearby (8 pc), quiescent mid-M dwarf (M 4.5V) of older age, made especially interesting by the hypothesized presence of a close-in planet invoked to explain its low-frequency radio emission. We analyzed the flaring activity of GJ 1151 to investigate its stellar atmosphere and magnetic properties using 2 minute short-cadence photometry from the Transiting Exoplanet Survey Satellite (TESS) and radio observations from the upgraded Giant Metrewave Radio Telescope (uGMRT). Analysis of TESS Sector 48 revealed three flare events, with bolometric flare energies estimated in the range of 10^{30}–10^{31} erg, showing that GJ 1151, previously regarded as optically inactive, does in fact exhibit flare activity, placing it in the lower-to-intermediate activity regime. Interestingly, we reported the first estimation of the most energetic flare energy to be $(4.86 \pm 0.19) \times 10^{31}$ erg that lasted for approximately 30 minutes, implying a minimum surface magnetic field strength of 45 G to power the event. We estimated the mean spot temperature to be approximately 2886 ± 837 K and the spot area to be 1.2% of the area of the star. The nondetection of radio emission at 1.36 GHz with the uGMRT suggests either time variability in the emission or the presence of a frequency cutoff in the 3σ upper limit on its flux density of 105 μJy beam^{-1} in the emission mechanism, potentially governed by the changing magnetic environment of the star and the orbital configuration of its hypothesized planet.

Unified Astronomy Thesaurus concepts: M dwarf stars (982); Stellar flares (1603); Starspots (1572); Magnetic fields (994); Habitable planets (695); Radio sources (1358)

1. Introduction

Planets in close-in orbits are immersed in the magnetized stellar wind generated by the expanding stellar corona. As they move through this wind, short-period planets can perturb its flow, transferring significant energy to the host star via sub-Alfvénic interactions (J. Saur et al. 2013). This energy can heat the star's chromosphere, creating a hot spot that varies with the planet's orbit. Such chromospheric variability is observed in hot Jupiter systems, pointing to star–planet interactions (SPIs; E. Shkolnik et al. 2008; P. W. Cauley et al. 2019). Similar SPIs have also been proposed for M dwarf systems with close-in rocky planets, such as TRAPPIST-1 (C. Fischer & J. Saur 2019). M dwarfs are the lowest-mass stars on the main sequence (0.08–0.55 $M_\odot$, 2400–3800 K; M. J. Pecaut & E. E. Mamajek 2013), the most abundant and long-lived stellar type objects, comprising ∼75% of the Milky Way's population (T. J. Henry 2007; J. J. Bochanski et al. 2010). Stellar flares in M dwarfs, similar to solar flares, are electromagnetic outbursts in the stellar corona, lasting from seconds to hours, with energies ranging from 10^{23} erg for nanoflares (e.g., C. E. Parnell & P. E. Jupp 2000) to 10^{33}–10^{38} erg for superflares (e.g., T. Shibayama et al. 2013). They serve as visible signatures of released magnetic energy through the reconnection of magnetic field lines (T. G. Forbes & L. W. Acton 1996). Most M dwarfs are found to be planet hosting (C. D. Dressing & D. Charbonneau 2015; M. Gillon et al. 2017). Although M dwarf activities such as spots, flares, and rotational variability (S. L. Hawley et al. 2014; J. Villadsen & G. Hallinan 2019) can erode planetary atmospheres, it also critically shapes the conditions of close-in habitable zones. Since most rocky exoplanets orbit M dwarfs, understanding their activity is essential to evaluate true habitability (V. S. Airapetian et al. 2017; S. Ranjan et al. 2017; H. K. Vedantham et al. 2020).

GJ 1151 (TIC 11893637), located 8 pc away from Earth, is a quiescent, slow-rotating M dwarf of spectral type (SpT) M4.5V (S. L. Hawley et al. 1996). It is a chromospherically less active star (E. R. Newton et al. 2017). The important properties of this star are listed in Table 1. The magnetic field is detected in GJ 1151 to be <680 G based on CARMENS data (A. Reiners et al. 2022). G. Foster et al. (2020) reported a small X-ray flare from GJ 1151 and detected a low coronal temperature of 1.6 MK, supporting the notion of minimal magnetic activity. H. K. Vedantham et al. (2020) detected radio emission from GJ 1151 using LOFAR Two-meter Sky Survey (LoTSS) data, reporting time-averaged Stokes I and V flux densities of 0.89 and 0.57 mJy over 8 hr between 120 and 170 MHz. The emission is strongly circularly polarized (64% ± 6%) and spectrally flat, resembling the plateau-like radio profiles observed from planets in our solar system (P. Zarka 1992). GJ 1151 was observed three additional times by LoTSS (each for 8 hr), but no emission was detected during those epochs. H. K. Vedantham et al. (2020) concluded that the LOFAR emission originates from GJ 1151 itself, not from a background source, based on its variability, strong circular polarization, and position. The author suggested that the radio emission originates in the stellar magnetosphere and is powered by the host star's magnetic field, with the orbiting planet acting as a trigger, analogous to the Jupiter–Io interaction (J. Saur et al. 2013).

Table 1
Stellar Parameters of GJ 1151

Parameters	Values	References
GJ	1151	(1)
TIC	11893637	(2)
R.A. (epoch J2000)	11 50 57.72	(3)
Decl. (epoch J2000)	+48 22 38.56	(3)
d (pc)	8.0426 ± 0.0036	(3)
G (mag)	11.6847 ± 0.0028	(3)
T (mag)	11.51	(2)
Spectral type	M4.5 V	(4)
Mass ($M_\odot$)	0.1639 ± 0.0093	(5)
Radius ($R_\odot$)	0.1781 ± 0.0042	(5)
$T_{\rm eff}$ (K)	3280 ± 40	(6)
log g (cgs)	5.09 ± 0.09	(6)
[Fe/H] (dex)	−0.12 ± 0.10	(6)
B (G)	<680	(7)
$P_{\rm rot}$ (d)	140 ± 10	(5)
$v\sin i$ (km s^{-1})	<2.0	(8)
Age (Gyr)	$6.1718^{+0.08914}_{-0.07358}$	(9)

References. (1) W. Gliese & H. Jahreiß (1979); (2) K. G. Stassun et al. (2019); (3) Gaia Collaboration et al. (2021); (4) S. L. Hawley et al. (1996); (5) J. Blanco-Pozo et al. (2023); (6) E. Marfil et al. (2021); (7) A. Reiners et al. (2022); (8) S. V. Jeffers et al. (2018); (9) Y. Lu et al. (2024).

The LOFAR emission from GJ 1151 closely aligns with theoretical models of interactions between an M dwarf and an Earth-sized planet in a 1–5 day orbit (H. K. Vedantham et al. 2020). The radial velocity (RV) follow-up studies (B. J. S. Pope et al. 2020; S. Mahadevan et al. 2021; M. Perger et al. 2021) found no evidence of such a planet. Radio emission models suggest the presence of an Earth-mass companion ($\leqslant 1.2 M_\oplus$) at short orbital separation (H. K. Vedantham et al. 2020), while subsequent RV measurements indicate a possible super-Earth with a minimum mass of $2.5 M_\oplus$ at 0.017 au (S. Mahadevan et al. 2021). This range of estimates highlights ongoing uncertainty, like the companion, which must be considered when interpreting the system's emission properties. Likewise, transit observations by B. J. S. Pope et al. (2021) failed to detect any close-in planetary signatures. B. J. S. Pope et al. also reported no significant flaring in the Transiting Exoplanet Survey Satellite (TESS) data of Sector 22, suggesting that GJ 1151 is optically inactive (B. J. S. Pope et al. 2021). In this study, our detection of flares in TESS Sector 48 is interesting in part because it contradicts the idea that GJ 1151 is an optically inactive star, thereby strengthening the case that its reported radio emission is more plausibly due to intrinsic stellar magnetic activity rather than SPI. J. Blanco-Pozo et al. (2023) detected a long-period planet ($>10.6 M_\oplus$, 390 day orbit) with flux too faint for detection ($\leqslant$0.1 mJy). They also found hints of another companion or stellar magnetic variability but could not rule out a low-mass, short-period planet or interactions involving an exomoon or local plasma as the source of the observed radio emission. Recently, M. Narang et al. (2024) reported nondetections of radio emission from GJ 1151 using upgraded Giant Metrewave Radio Telescope (uGMRT) observations at 150, 218, and 400 MHz and suggested that the emission is highly time variable and likely modulated by the star–planet system's orbital phase and the host star's magnetic field. J. R. Callingham et al. (2021) detected coherent radio emission from 19 M dwarfs using LoTSS data, suggesting such emission is common among main-sequence M dwarfs, with some likely due to plasma processes, though emission from quiescent, slow rotators likely involves additional mechanisms such as SPI.

GJ 1151 is a quiescent magnetically active star; thus, we aim to study the magnetic activities based on its flaring activities. We investigate the flaring activities of GJ 1151 using the 2 minute short-cadence TESS photometric light curves. Previous studies investigated the radio emission from GJ 1151 across the 120–400 MHz frequency range. High-frequency (1.12–1.62 GHz) radio emission has been detected from another M dwarf AD Leo (R. A. Osten & T. S. Bastian 2006). More recently, GHz-frequency electron cyclotron maser signals have been reported in other nearby M dwarfs, with a possible SPI as the driver, including Proxima Centauri (M. Pérez-Torres et al. 2021), YZ Ceti (J. S. Pineda & J. Villadsen 2023), and GJ 3323 (K. N. Ortiz Ceballos et al. 2024). Hence, to explore and quantify the radio emission and investigate its nature at higher frequencies (>400 MHz), we observed the GJ 1151 system at another frequency band: 1.36 GHz (using band 5 of uGMRT). This paper is organized as follows: we describe the observations and data reduction in Section 2, we reported our obtained results in Section 3, and we discuss our obtained results in Section 4. Finally, we have summarized our work in Section 5.

2. Observations and Data Reduction

2.1. Photometry

The TESS is a space-based telescope developed under NASA's Explorer program and was launched on 2018 April 18. Its primary mission is to detect transiting exoplanets around bright stars across the sky. TESS is equipped with four wide-field cameras, each with a 10.5 cm aperture, operating in the red bandpass range of 600–1000 nm. Together, they continuously monitor a $24° \times 96°$ strip of the sky for 27 consecutive days (see G. R. Ricker et al. 2014 for details).

GJ 1151 (TIC 11893637, $T_{\rm mag} = 11.51$ K. G. Stassun et al. 2019) was observed in three of the TESS sectors, in Sector 22 (2020 February 19–2020 March 17), Sector 48 (2022 January 28–2022 February 25), and Sector 75 (2024 January 30–2024 February 26), respectively. We have selected a 2 minute cadence of TESS Sector 22 data in two orbits (Orbit 51 and 52) as a part of its Cycle 2 observations, Sector 48 data in two orbits (Orbit 103 and 104) as a part of its Cycle 4 observations, and Sector 75 data in two orbits (Orbit 157 and 158) as a part of its Cycle 6 observations. First, we used the "lightkurve" (Lightkurve Collaboration et al. 2018) package to download the TESS light curves of our target for three sectors, 22, 48, and 75, from the Mikulski Archive for Space Telescopes archive. Then, the Science Processing Operations Center (SPOC) pipeline (J. M. Jenkins et al. 2016) extracts the 2 minute cadence data. We have chosen the Presearch Data Conditioning (PDCSAP) light curves in our analysis (M. C. Stumpe et al. 2012; J. C. Smith et al. 2012), which are Simple Aperture Photometry flux (SAP FLUX) after removing the long-term trends using cotrending basis vectors. The "presearch data conditioning" (PDC) module also recognizes and records outliers and makes corrections for the crowding contamination from surrounding nearby stars, where SAP FLUX does not (N. M. Guerrero et al. 2021). We have taken only the data with QUALITY = 0, while the nonzero value of QUALITY indicates that the data have been compromised to some degree by instrumental effects. We also removed the outliers and not a number (NaNs) and normalized

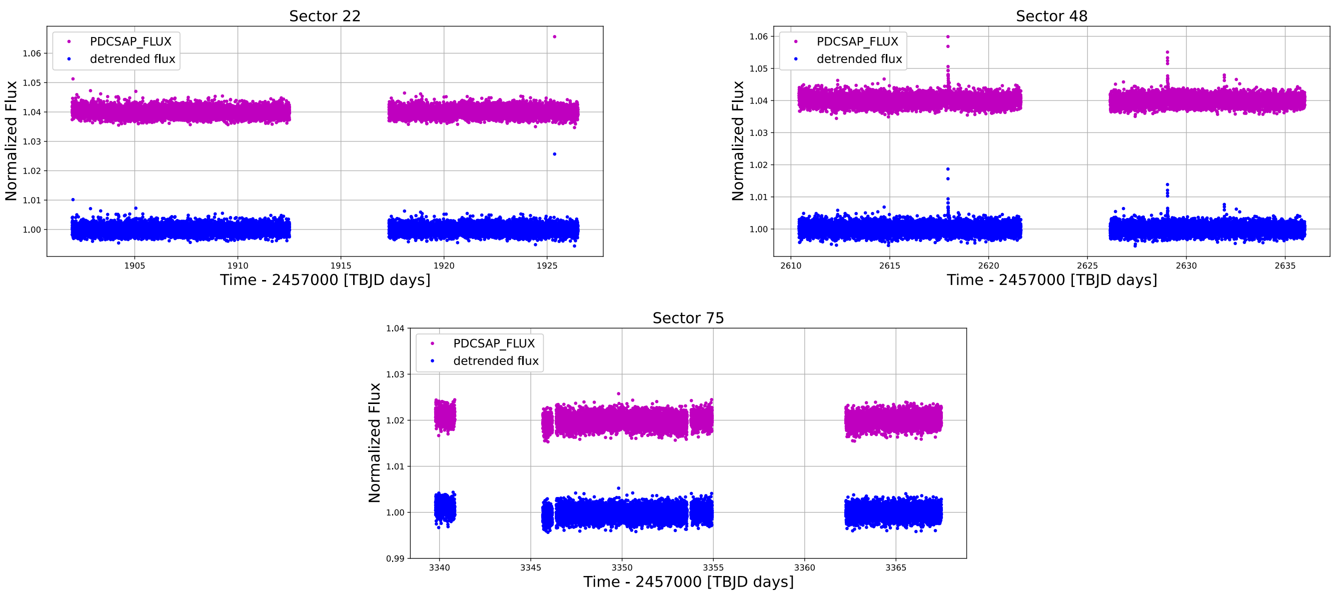

Figure 1. TESS PDCSAP light curve (pink) and detrended light curve (blue) of GJ 1151 in Sector 22 (upper left), Sector 48 (upper right), and Sector 75 (bottom). The detrending procedure is described in Section 3.1.

Table 2
Log of Radio Observation of GJ 1151 on uGMRT

Date	Start Time (UTC)	End Time (UTC)	Central Observing Frequency (GHz)	Total Time Spent On-source (minutes)	Flux Calibrator	Phase Calibrator
2022-12-12	02:00	04:06	1.36	101	3C 147	1145 + 497

each light curve by dividing each point's flux by the target mean flux. The SPOC pipeline corrects for the dilution from other nearby stars in and around the TESS aperture, as denoted by the CROWDSAP value in the TESS header files. The values of CROWDSAP are 0.996, 0.997, and 0.998 in three TESS Sectors 22, 48, and 75, respectively. Therefore, there is no potential effect of nearby bright stars. The PDCSAP FLUX light curves are shown in Figure 1.

2.2. Radio Observation Using the uGMRT

Observations of GJ 1151 were conducted with the uGMRT[4] on 2022 December 12 as part of proposal ID 43_113 (PI: Diya Ram). The source was observed in Band 5 (1.05–1.45 GHz), with a total on-source time of approximately 2 hr centered at 1.36 GHz. The flux density scale was set using 3C 147, observed twice—once at the start and once at the end of the observing session. Phase calibration was performed using 1145+497, which was observed in a regular cycle of $\sim$25 minutes on the target source and $\sim$5 minutes on the phase calibrator. Further details of the observations are provided in Table 2.

The Astronomical Image Processing System[5] software package was used for reduction of the data. The R. A. Perley & B. J. Butler (2013) flux density scale was used. Time- and channel-based flagging was performed to mitigate data affected by radio frequency interference. Additionally, baselines exhibiting large closure errors on the bandpass calibrator were identified and removed. Antennas showing high errors in antenna-based solutions during certain time intervals were examined, and their data were flagged as necessary. The final image was produced after multiple rounds of phase-only self-calibration, followed by a single round of amplitude and phase self-calibration. For the imaging and cleaning, we have used the task IMAGR and made the final image of the target source.

3. Results

3.1. Identification of Stellar Flare in TESS

An open-source Python software ALTAIPONY[6] (E. Ilin et al. 2021) is used to detect the flare events from the TESS light curves and estimate the flare parameters of each event. Initially, we applied a detrending approach to remove astrophysical trend-like modulation due to the starspots and instrumental trends from the time-series data. To detrend the PDCSAP light curves, we fit and subtract a third-order spline function that goes through the start and end of any light-curve portion with no gaps longer than 2 hr. Then, the strong sinusoidal signal is iteratively removed from the light curves. This iteration process first masks outlier points using the σ-clipping method. When the light curve showed a single outlier above 3σ, it treated the data point as a pure outlier. At the same time, when a series of data points exceeded 3σ, they were considered flare candidates. Then we calculate a Lomb–Scargle periodogram for the light curve and perform a least-squares fit with a cosine function using the dominant frequency in the periodogram as a starting point. The light curve was then subtracted from this cosine fit, and this procedure continued until the dominant peak's signal-to-noise

4 http://gmrt.ncra.tifr.res.in
5 http://www.aips.nrao.edu/
6 https://altaipony.readthedocs.io/en/latest/

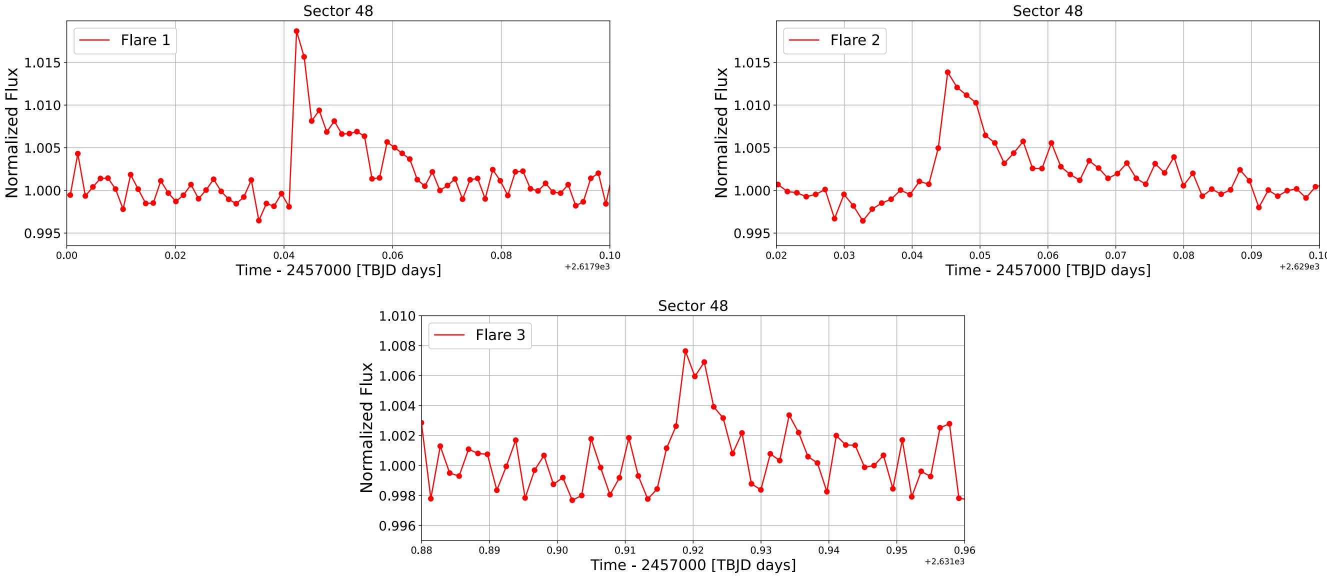

Figure 2. Flare events of GJ 1151 in TESS light curve.

ratio decreased to 1. Finally, any nonsinusoidal variability was eliminated using two third-order *Savitzky–Golay* filters ("*savgol*" package; A. Savitzky & M. J. E. Golay 1964) with a window size of 6 and 3 hr.

To identify the flare events, we used the task *find_flares()* under the ALTAIPONY (E. Ilin et al. 2021). This process involved using three parameters, namely, N1, N2, and N3, with values set to 3, 2, and 3, respectively, establishing the criteria for flare event detection (S. W. Chang et al. 2015). Following detrending and flare identification in ALTAIPONY, a visual inspection of all flare light curves was conducted to check for any inconsistency. The key characteristics considered were a distinct sharp rise and slow exponential decay. Furthermore, we again zoomed in on the particular flare region and checked visually that three consecutive data points surpass the 3σ threshold of the light curve and be spared at least 6 minutes above it. We have identified three flaring events in the 2 minute short cadence of the TESS light curve in Sector 48 only. We could not find any flaring events from the TESS light curves in Sectors 22 (B. J. S. Pope et al. 2021) and 75. If the flare rate were constant across equal TESS sectors, the probability of observing the sequence (0, 3, 0) is only 0.83% based on Poisson distribution. A χ^2 test ($\chi^2 = 6.0$, $p = 0.05$) further indicates statistical evidence that GJ 1151's flare rate is variable. H. K. Vedantham et al. (2020) reported radio emission from GJ 1151 over an 8 hr interval between 120 and 170 MHz using LOFAR, whereas M. Narang et al. (2024) found no detection in 33 hr of uGMRT observations at 150, 218, and 400 MHz. Magnetic field estimates for GJ 1151 range from ∼26 to 100 G (H. K. Vedantham et al. 2020; M. Narang et al. 2024). This contrast suggests that flare-rate variability, as seen in TESS, is consistent with the episodic nature of past radio detections and corresponding magnetic field constraints. The observed flare events are shown in Figure 2.

3.2. *Estimation of Stellar Flare Energy*

We initially estimated the equivalent duration (ED) of a flare to calculate its energy. It is defined as the equivalent time it takes for a star to release the same amount of energy at the quiescent stage as it releases during a flare (R. E. Gershberg 1972). It is measured as the area under the flare light curve in the unit of time. To calculate ED, we implemented the open-source Python software ALTAIPONY (E. Ilin et al. 2021), which automatically detects and characterizes the flares in the object. Further analysis using ALTAIPONY provides several flare parameters, e.g., start and stop time, flare amplitude, ED, etc. Then, the ED was used to calculate the energy of flares. To calculate the flare energy E_{flare}, we generated the synthetic photospheric spectrum of the object by using BT-Settl model.[7] The fundamental parameters of GJ 1151 like the effective temperature (3280 ± 40 K; E. Marfil et al. 2021), surface gravity (5.09 ± 0.09 dex; E. Marfil et al. 2021), and metallicity (−0.12 ± 0.10 dex; E. Marfil et al. 2021) are used to generate the synthetic spectra. J. A. G. Jackman et al. (2023) estimated a flare temperature of 10,700 K for fully convective M stars after accounting for UV line emission. For bolometric energies of white-light flares in TESS, we follow T. Shibayama et al. (2013), assuming a 9000 K blackbody spectrum, consistent with our earlier works (D. Ram et al. 2025a, 2025b). Since GJ 1151 is not fully convective but has a radiative core, the 9000 K assumption is more appropriate here. M. Kretzschmar (2011) found that the blackbody spectrum spectral energy distribution of flares can be fitted well with that temperature. Following T. Shibayama et al. (2013), we estimated the bolometric flare luminosity L_{flare} using the temperature T_{flare} and area A_{flare} of the flare by the Stefan–Boltzmann law:

$$L_{\mathrm{flare}} = \sigma_{\mathrm{SB}} T_{\mathrm{flare}}^4 A_{\mathrm{flare}}, \tag{1}$$

where σ_{SB} is the Stefan–Boltzmann constant. The area of the flare is estimated as follows:

$$A_{\mathrm{flare}} = \frac{(F_i - F_o)}{F_o} \pi R_{\mathrm{star}}^2 \frac{\int R_\lambda F_{\lambda(T_{\mathrm{eff}})} d\lambda}{\int R_\lambda B_{\lambda(T_{\mathrm{flare}})} d\lambda}, \tag{2}$$

where λ is the wavelength, $B_{\lambda(T)}$ is the Planck function with each wavelength λ, $F_{\lambda(T)}$ is the flux of the synthetic photosphere spectrum, and R_λ is the response function of the

[7] http://svo2.cab.inta-csic.es/theory/newov2/

Table 3
Results of Flare Analysis of GJ 1151 in TESS Sector 48 in This Work

t_s (TBJD-2457000) (Time)	t_f (TBJD-2457000) (Time)	Rel. Flux a	Duration (minutes)	ED (s)	$E_{\rm bol}^{\rm syn}$ (erg)	Magnetic Field (Gauss)
2617.9423	2617.9631	0.018	30.00	13.26 ± 0.53	$4.86 \pm 0.19 \times 10^{31}$	45
2629.0438	2629.0535	0.013	13.99	7.76 ± 0.40	$2.84 \pm 0.14 \times 10^{31}$	34
2631.9188	2631.9230	0.007	6.00	2.48 ± 0.27	$9.10 \pm 0.98 \times 10^{30}$	19

Note. t_s: start time of flare. t_f: stop time of flare. a: relative amplitude of flare. Duration: duration of flare. ED: equivalent duration of flare. $E_{\rm bol}^{\rm syn}$: bolometric energy of flare by taking the synthetic spectra of the object.

TESS bandpass. The flare amplitude is defined as $\Delta F/F = (F_i - F_o)/F_o$, where F_i is the maximum flux at the flare and F_o is the local flux of a quiescent star (S. L. Hawley et al. 2014). Then, the total bolometric energy of the flare ($E_{\rm flare}$) is an integral of $L_{\rm flare}$ during the flare duration (T. Shibayama et al. 2013):

$$E_{\rm flare} = \int L_{\rm flare}(t)dt = \sigma_{\rm SB}\pi R_{\rm star}^2 T_{\rm flare}^4 \times \frac{\int R_\lambda F_{\lambda(T_{\rm eff})}d\lambda}{\int R_\lambda B_{\lambda(T_{\rm flare})}d\lambda} \int \left(\frac{F_i - F_0}{F_0}\right) dt. \quad (3)$$

The final time-integrated quantity is commonly referred to as the ED of the flare, which is computed by integrating the relative flux increase over the full duration of the flare event (S. L. Hawley et al. 2014; K. Ikuta et al. 2023; R. Kumbhakar et al. 2023; S. Ghosh et al. 2024; R. Kumbhakar et al. 2025).

We estimated the flare energy in the range of 10^{30}–10^{31} erg in the three identified flare events of TESS Sector 48 from Equation (3) (see Table 3). We calculated the uncertainty in the flare energy estimation using a Monte Carlo approach with 10,000 iterations of the parameters used to derive the bolometric energy. This method allowed us to robustly propagate the uncertainties in stellar parameters—specifically, the effective temperature, stellar radius, and ED—into the final flare energy estimate (J. A. G. Jackman et al. 2021). Using this approach, we report for the first time the bolometric energy of a highly energetic flare on GJ 1151, measured to be $(4.86 \pm 0.19) \times 10^{31}$ erg, with a total duration of approximately 30 minutes, as observed in TESS Sector 48.

3.3. Estimation of Magnetic Field from the Flare Events

Magnetic fields estimated from flares reveal the star's energy reservoir (K. Shibata & T. Magara 2011), trace dynamo evolution (A. Reiners et al. 2022), and help assess SPIs and atmospheric erosion, key for exoplanet habitability (M. Tuomi et al. 2019). Considering similarities between the physical processes involved in the solar flares and stellar flares, several authors obtained the lower limit of the maximum field strength $B_z^{\rm max}$ using scaling relations between the flare energy and magnetic field (G. Aulanier et al. 2013; R. R. Paudel et al. 2018; Y. Notsu et al. 2019). To estimate the magnetic field from the solar flares, G. Aulanier et al. (2013) assumed that a certain fraction (∼20%) of the total available magnetic energy is released by magnetic reconnection through an active region, and the active region is assumed to be associated with positive and negative polarity (bipolar spot) with a linear separation of $L^{\rm bipole}$. Following the relation given by G. Aulanier et al. (2013) and R. R. Paudel et al. (2018), we can estimate the lower limit of the magnetic field strength ($B_z^{\rm max}$) to produce such flares:

$$E_{\rm flare} = 0.5 \times 10^{32} \left(\frac{B_z^{\rm max}}{1000\ {\rm G}}\right)^2 \left(\frac{L^{\rm bipole}}{50\ {\rm Mm}}\right)^3 {\rm erg}, \quad (4)$$

where $E_{\rm flare}$ is the bolometric flare energy and $L^{\rm bipole}$ is the linear separation between bipoles. We consider upper limit of bipole length, $L^{\rm bipole}$ equal to πR as adopted by previous authors (G. Aulanier et al. 2013; R. R. Paudel et al. 2018). We consider R, the radius of GJ 1151, to be $0.1781 \pm 0.0042 R_\odot$ (J. Blanco-Pozo et al. 2023). We have detected a total of three flare events in Sector 48 having flaring energy in the range of 10^{30}–10^{31} erg. We estimated the lower limit of the magnetic field strength ($B_z^{\rm max}$) from Equation (4) for such flare energies in the range of 19–45 G in Sectors 48 (see Table 3).

3.4. Area of the Starspot

The area of starspots could be determined in several ways, e.g., through Zeeman–Doppler Imaging (L. Rosén et al. 2015), spectral modeling (M. A. Gully-Santiago et al. 2017), planet-transit spot modeling (B. M. Morris et al. 2017), etc. The size of the spot could also be estimated from the amplitude and rotational modulation of the light curves (L. M. Rebull et al. 2016; H. A. C. Giles et al. 2017). Following the method of Y. Notsu et al. (2013), originally developed for solar-type stars but reasonably extendable to M dwarfs due to their shared dynamo-driven magnetic activity and spot formation, we estimate the area of the spot using the following relations:

$$T_{\rm star} - T_{\rm spot} = 3.58 \times 10^{-5} T_{\rm star}^2 + 0.249 T_{\rm star} - 808, \quad (5)$$

$$A_{\rm spot} = A_{\rm star}\left(\frac{\Delta F}{F}\right)\left[1 - \left(\frac{T_{\rm spot}}{T_{\rm star}}\right)^4\right]^{-1}, \quad (6)$$

where $T_{\rm star}$ is the effective photospheric temperature and $T_{\rm spot}$ is the spot temperature. The $\Delta F/F$ is the amplitude of the normalized TESS light curve. We estimate the amplitude of GJ 1151 to be 1.00023 in TESS data by phasing the light curves from combined sectors and taking the maximum value of the phased mean light curve, without considering the flares measured from the phase-folded and binned light curves (see Figure 3). $A_{\rm star}$ is the area of the stellar disk calculated as $2\pi R^2$ (R is the radius of the star, $0.1781 \pm 0.0042\, R_\odot$; J. Blanco-Pozo et al. 2023). Considering the effective photospheric temperature of GJ 1151 of 3280 ± 40 K (E. Marfil et al. 2021), we have estimated $T_{\rm spot}$ to be 2886 ± 837 K using Equation (5). Using Equation (6), we estimated the approximate spot area to be 1.2% of the total surface area of GJ 1151.

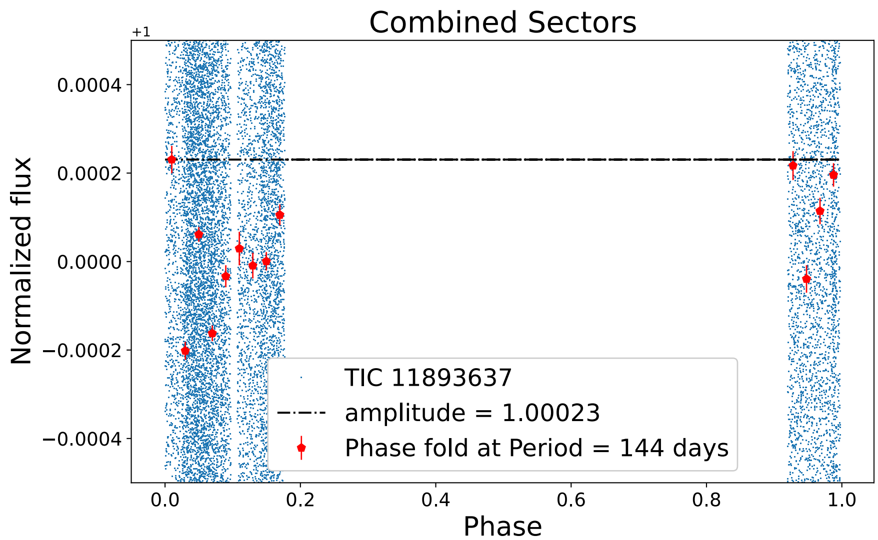

Figure 3. The phased light curve of GJ 1151 from combined sectors of the TESS observations (blue dots); the red dots indicate the binned light curve, and the dashed-dotted line indicates estimated maximal amplitude 1.00023 of GJ 1151.

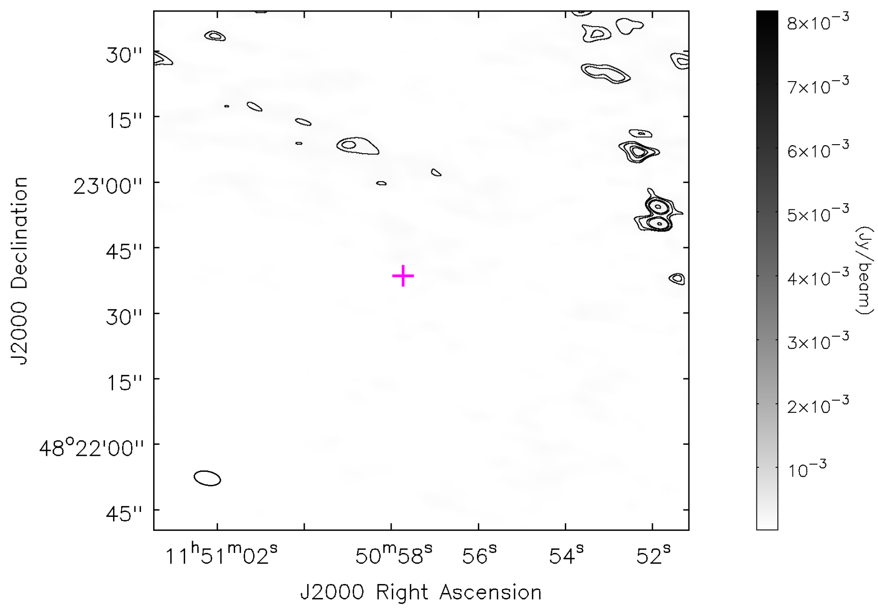

Figure 4. Radio image of GJ 1151 on 2022 December 12 at an observing frequency of 1.36 GHz from the legacy uGMRT. The position of the object is marked with the plus sign. The source is not detected at band 5. The rms of the image is 35 μJy beam^{-1}. The contour of the image is drawn with a 3σ limit, where σ is the local rms noise.

3.5. Nondetection of Radio Emission in GJ 1151 at 1.36 GHz Using uGMRT

We analyzed the uGMRT data at 1.36 GHz of GJ 1151, observed on 2022 December 12 (see Table 2). In band 5 at 1.36 GHz, we achieve an rms sensitivity of 35 μJy beam^{-1}. The 3σ upper limit of the flux density is estimated to be 105 μJy. This is much smaller than the flux of 0.89 mJy detected by LOFAR (H. K. Vedantham et al. 2020). No radio emission was detected from the source at 1.36 GHz (see Figure 4). A possible reason for this nondetection is described in Section 4.2.

4. Discussion

GJ 1151 is a quiescent, slow-rotating magnetically active star (period of the star, 140 ± 10 days; J. Blanco-Pozo et al. 2023). From the TESS light curves of GJ 1151, we found bolometric flare energies of 10^{30}–10^{31} erg; these flare energies are consistent with those of slowly rotating M dwarfs (10^{30}–10^{32} erg; J. R. A. Davenport 2016; E. R. Newton et al. 2017), placing GJ 1151 in the lower-to-intermediate activity regime. Flare-derived magnetic fields in the range of 19–45 G reveal stellar energy, trace dynamo evolution, and constrain SPIs and atmospheric erosion critical for habitability

(K. Shibata & T. Magara 2011; M. Tuomi et al. 2019; A. Reiners et al. 2022). The measured spot area, estimated to be 1.2% of the area of the star, and the mean spot temperature, estimated to be approximately 2886 ± 837 K, set the stellar magnetic energy budget, shaping flare-driven UV/X-ray flux that impacts planetary atmospheres and habitability (V. S. Airapetian et al. 2020; M. Aizawa et al. 2022).

4.1. Flares in GJ 1151: Habitability in the Probable Planet?

GJ 1151 hosts at least one confirmed exoplanet, GJ 1151 c, a Neptune-like planet with a minimum mass of approximately $10.6\,M_\oplus$ and an orbital period of ∼390 days at a distance of 0.57 au (J. Blanco-Pozo et al. 2023). An earlier candidate, GJ 1151 b, was proposed based on low-frequency radio emissions attributed to star–planet magnetic interactions (H. K. Vedantham et al. 2020), suggesting a close-in terrestrial planet with an orbital period of ∼2 days with a minimum mass of approximately $2.5\,M_\oplus$ at a distance of 0.017 au (S. Mahadevan et al. 2021). However, this inner planet remains unconfirmed due to a lack of supporting RV evidence in follow-up observations (B. J. S. Pope et al. 2020; M. Perger et al. 2021).

The magnetic field reaches Earth at a distance of $\sim$6–$9R_\oplus$ from the Sun's side. Although GJ 1151 is less active than the Sun and likely has a weaker stellar wind, the star–planet separation dominates magnetospheric compression; at 0.017 au (the putative 2 day orbit), the greatly enhanced wind ram pressure would still strongly compress the planetary magnetosphere for the same magnetic field. Assuming a dipolar stellar field inside or near the Alfvén surface about 20% along the planetary orbital path (A. A. Vidotto et al. 2014), the stellar field at orbital distance can be expressed as (S. Bharati Das et al. 2019)

$$B = B_* \times \left(\frac{R_*}{r}\right)^3. \quad (7)$$

Here, B_* (<680 G; A. Reiners et al. 2022) is the surface magnetic field of GJ 1151, R_* ($0.17\,R_\odot$; J. Blanco-Pozo et al. 2023) is the radius of star, and r (0.57 au for GJ 1151 c; J. Blanco-Pozo et al. 2023 and 0.017 au for GJ 1151 b; S. Mahadevan et al. 2021) is the distance between GJ 1151 and its orbiting planet. Using Equation (7), we estimate that the planet's magnetic dipole field strength of 2.09×10^{-6} G would be required for SPIs to occur at the orbital distance of GJ 1151 c (0.57 au), whereas a significantly stronger field of 0.06 G would be necessary for interactions at 0.017 au with the orbit of the candidate planet GJ 1151 b. We estimate the magnetic field of 45 G from the 4.86×10^{31} erg flare observed in TESS Sector 48 represents a local coronal loop field near the stellar surface of GJ 1151, typically confined to heights of only a few stellar radii ($\lesssim$3–$4\,R_*$ M. J. Aschwanden 2005; A. F. Kowalski et al. 2013), and should not be interpreted as either the planetary magnetic field (B_p) or the stellar large-scale field at the planetary orbit. The putative 2 day planet lies at $a = 0.017$ au (J. Blanco-Pozo et al. 2023), i.e., roughly $16\,R_*$ beyond plausible flare loop apices, meaning the coronal flare field cannot directly extend to the planet. SPI is governed by the large-scale stellar field in orbit ($B_*(r)$), while the flare-derived field shows the magnetic energy reservoir of GJ 1151 capable of driving winds and disturbances that enhance the SPI signatures.

Following the relation of M. N. Günther et al. (2020), the flare frequency needed to drive prebiotic chemistry or abiogenesis zones (P. B. Rimmer et al. 2018; J. Xu et al. 2018) can be expressed as

$$\nu \geqslant 25.5\ \mathrm{day}^{-1}\left(\frac{10^{34}\ \mathrm{erg}}{E_U}\right)\left(\frac{R}{R_\odot}\right)^2\left(\frac{T_{\mathrm{star}}}{T_\odot}\right)^4. \quad (8)$$

Studying the stellar flares from a sample of M dwarfs, M. N. Günther et al. (2020) found that 7.6% of the flare's bolometric energy falls into the U band, leading to $E_U \approx 7.6\% E_{\mathrm{bol}}$, by using the U-band spectral response function, and the authors assume a 9000 K blackbody for the flare. As GJ 1151 is a similar spectral type (SpT) object, we can scale the above Equation (8) for our object. We estimate that the flare frequency needed to drive the prebiotic chemistry in the planet (GJ 1151 c) is ≈82 flares day^{-1} with the bolometric flare energy of $\sim 10^{31}$ erg from the above Equation (8). We estimated the flare frequency in the range of 0.14 flares day^{-1} with the bolometric flare energy of $\sim 10^{31}$ erg based on TESS data. This moderate but infrequent activity is insufficient to drive abiogenesis under current conditions, and repeated flaring is also unlikely to improve present-day habitability, since atmospheric erosion and ozone loss would hinder the survival of surface life (M. Tuomi et al. 2019). The slow rotation of 140 ± 10 days (J. Blanco-Pozo et al. 2023) suggests that GJ 1151 is an old mid-M dwarf (A. Reiners et al. 2022), consistent with its low flare frequency compared to highly active stars like AD Leo and GJ 1243 (S. L. Hawley & B. R. Pettersen 1991; S. L. Hawley et al. 2014). It is also less active than Proxima Centauri, which, despite being of similar age, shows frequent powerful flares (K. Vida et al. 2019). Similar studies on AD Leo, GJ 1243, Proxima Centauri, and TRAPPIST-1 conclude that while early intense flaring could have supported abiogenesis, sustained activity at later ages mainly poses a threat to long-term habitability (K. Vida et al. 2017; P. J. Wheatley et al. 2017; W. S. Howard et al. 2018).

4.2. Radio Signatures of Probable Star–Planet Interaction

The radio emission was detected from GJ 1151 using LOFAR over 8 hr between 120 and 170 MHz (H. K. Vedantham et al. 2020). The emission exhibits strong circular polarization (64% ± 6%) and a spectrally flat profile, closely resembling the plateau-like radio signatures observed in the solar system, which initially suggested a possible SPI can originate in a sub-Alfvénic interaction of its magnetospheric plasma with a short-period exoplanet (H. K. Vedantham et al. 2020). J. Blanco-Pozo et al. (2023) observed higher magnetic activity in the star, causing RV, activity index, and photometric variability, suggesting the expected radio emission from a hidden short-period planet interacting with the star's magnetic field. This SPI drives currents that produce auroral-type radio bursts via ECMI, not from the outer 390 day companion (GJ 1151c; H. K. Vedantham et al. 2020; S. Mahadevan et al. 2021; J. Blanco-Pozo et al. 2023). An emission frequency of ∼140 MHz implies a local magnetic field of at least 50 G (S. Mahadevan et al. 2021). M. Narang et al. (2023) showed that a ∼50 G magnetic field would require a planet mass of ∼1 M_J, far exceeding the estimated mass of GJ 1151. Zeeman–Doppler imaging by L. T. Lehmann et al. (2024) shows that the magnetic obliquity of GJ 1151 varies significantly and that the star was magnetically quiet during their observations. They estimated the average dipole field strength to be 26 and 62 G, placing the emission peak around 64–176 MHz. The flux at higher frequencies diminishes significantly, making our 1.36 GHz uGMRT observation too high to detect the emission, which may explain the nondetection during our observation. J. Blanco-Pozo et al. (2023) suggested that GJ 1151 may host a long-period (390 day) substellar companion. LOFAR detected strongly polarized radio emission between 120 and 170 MHz (H. K. Vedantham et al. 2020), and similar bursts are commonly observed from brown dwarfs (E. Berger 2006; E. Berger et al. 2008; G. Hallinan et al. 2008; M. Route & A. Wolszczan 2016). Therefore, a long-period substellar companion could plausibly be the source of the LOFAR emission rather than the planet GJ 1151 c or the star itself (M. Narang et al. 2023, 2024). A JWST Cycle 3 program (PID 5497; G. Stefansson et al. 2024) has been approved to search for a potential companion around GJ 1151 using NIRCam coronography with F200W and F444W filters, enabling the detection of companions as small as 3 M_J at separations between 3 and 25 au.

5. Summary and Conclusion

We present here the results of the light-curve analysis and flaring activities on GJ 1151 using TESS 2 minute short-cadence photometric data in Sectors 22, 48, and 75, respectively, and the radio data using uGMRT band 5 data at 1.36 GHz.

1. From these short-cadence TESS data, we found three flare events in Sector 48, while we could not find any flare events in the other two sectors, 22 and 75, respectively. Interestingly, we, for the first time, estimate a high-flare event of GJ 1151 of the bolometric flare energy to be $(4.86 \pm 0.19) \times 10^{31}$ erg with a duration of 30 minutes.
2. We estimated the bolometric flare energy in the range of 10^{30}–10^{31} erg. Using the bolometric flare energy based on the TESS flare events, we estimated the lower limit of the magnetic field in the range of 19–45 G. Such flaring activities depend on the size and distribution of the area of starspots, which is estimated to be 1.2% of the area of the star and the mean spot temperature of approximately 2886 ± 837 K.
3. We analyzed the uGMRT data at 1.36 GHz of GJ 1151, achieving an rms sensitivity of 35 μJy beam^{-1}. No radio emission was detected from the source at 1.36 GHz due to low sensitivity, indicating that the radio emission is either time variability in the emission or the presence of a frequency cutoff in the 3σ upper limit on its flux density of 105 μJy beam^{-1} in the emission mechanism, probably modulated by the orbital phase of the star–planet system and the magnetic field of the host star.

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Data Availability

All the TESS data used in this paper can be found in MAST: doi:10.17909/ps77-ya91. All the uGMRT data used in this paper can be found in the GMRT online archive: https://naps.ncra.tifr.res.in/goa/data/search.

Facilities: TESS and uGMRT.

Software: astropy (Astropy Collaboration et al. 2013, 2018), astroquery (A. Ginsburg et al. 2019), lightkurve (Lightkurve Collaboration et al. 2018), AltaiPony (J. R. A. Davenport 2016; E. Ilin et al. 2021), numpy (C. R. Harris et al. 2020), matplotlib (J. D. Hunter 2007), AIPS (E. W. Greisen 1990).

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